

High-Frequency Broadband RF Transmit-Receive Switch for Pulsed Magnetic Field NMR

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Abstract—The utility of high-field, high-frequency broadband pulsed magnetic field nuclear magnetic resonance (PMF-NMR) is significant in matter physics. The radio frequency (RF) transmit-receive (TR) switch is a kernel device used in nuclear magnetic resonance (NMR) spectrometers to switch signals between the two states of transmitting high-power RF pulses and receiving weak NMR signals. Most designs currently used in NMR have the disadvantages of a narrowband characteristic, low power capacity or large insertion loss, which make them unsuitable for high-frequency broadband PMF-NMR experiments. To improve the performance TR switches in PMF-NMR, a new design of an approximately symmetrical TR switch topology based on PIN diodes and multiple sections of coplanar waveguide lines with different impedances is introduced in this article. By setting the parameters of the impedance lines and the values of the inductors and capacitors as variables and using the gradient optimization and the least square error evaluation function, we obtain excellent parameters for achieving the high-frequency broadband TR switch. The prototype of an active TR switch is designed and implemented for PMF-NMR with a band of 250 MHz–1 GHz, an insertion loss that is less than 0.56 dB, a switching time that is less than 0.9 μ s and a high-power capacity of 200 W.

Index Terms—High-frequency broadband transmit-receive (TR) switch, nuclear magnetic resonance (NMR), PIN diode, pulsed magnetic field (PMF), radio frequency (RF).

I. INTRODUCTION

NUCLEAR magnetic resonance (NMR) is an important means to study the microstructure of matter in condensed matter physics. In NMR experiments, the increase in the field strength can improve the resolution of the spectra and the signal-to-noise ratio (SNR) [1]. In recent decades, pulsed magnetic field NMR (PMF-NMR) has been proposed and has been used in high-temperature superconductors, frustrated spin-1/2 chain compounds and so on, which have special physical properties. PMF-NMR experiments of nuclei such as ^{63}Cu , ^{65}Cu , and ^{51}V in $\text{YBa}_2\text{Cu}_3\text{O}_{6.51}$ and LiCuVO_4 have

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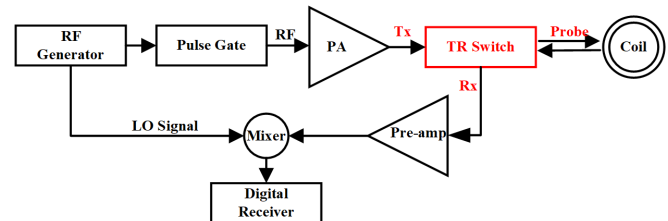


Fig. 1. Simplified diagram of the RF transceiver subsystem in a typical NMR spectrometer.

raised the resonance frequency to more than 500 MHz and discovered or verified the special physical properties, which can only be detected under ultrahigh field strengths [2], [3]. Kozlov et al. [4] even achieved the 2.4 GHz resonant frequency of ^1H under the 56 T PMF.

A Simplified diagram of the radio frequency (RF) transceiver subsystem in a typical NMR spectrometer is shown in Fig. 1. An RF generator provides two RF signals of the same frequency. One is used as the input of the pulse gate, and the other is used as the local oscillator (LO) signal. During the transmit state, the high-power RF signal output by the power amplifier (PA) is transmitted to the coil through the transmit-receive (TR) switch. In this state, it is necessary to ensure that the high-power RF signal is transmitted from the Tx port of the TR switch to the Probe port with the minimum insertion loss, and ensure that there is sufficient isolation between the Tx port and the Rx port to prevent damage to the preamplifier (Pre-amp). In the receiving state, the PA is blanked out and the Probe port and the Rx port of the TR switch are connected. The weak NMR signal is transmitted to the Pre-amp through the TR switch, and then input to the digital receiver after being down-converted with the LO signal in the mixer. The TR switch plays a kernel role in time-multiplexing the transmitted and received signals, and therefore, its characteristics can affect the performance of the entire system.

In condensed matter physics NMR experiments, only a single coil is generally used to apply high-power excitation to the samples and detect the NMR signals. The rapid switching of the two working states is completed by the TR switch [5]. Among the existing RF TR switches, there are passive and active switches that achieve switch-state transitions. As Fig. 2 shows, the conventional passive switches use a $\lambda/4$ line and RF diodes to passively achieve the switching of transmission and reception [6]. However, passive switches have narrow frequency bands and a low isolation due to the narrowband characteristics of $\lambda/4$ lines, and the nonideal properties of

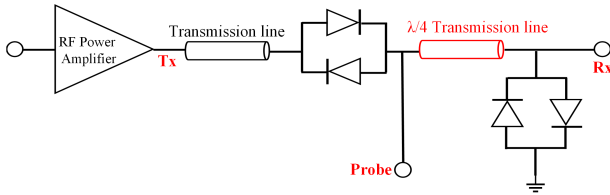


Fig. 2. Schematic of a conventional passive TR switch using a $\lambda/4$ line and cross diodes.

diodes, such as the resistance is not equal to 0 during conduction. For a fixed length of the $\lambda/4$ line, the frequency band range of the impedance transformation is only within $\pm 10\%$ of its center frequency [7]. Therefore, the bandwidth of a passive switch with a fixed $\lambda/4$ line generally cannot exceed 100 MHz.

Compared with steady-state field NMR, PMF-NMR has a higher background field strength, which causes it to have a higher resonance frequency and a broadband characteristic caused by the increased resonance frequency difference between different nuclei. Therefore, PMF-NMR is usually used for single-nucleus, single-frequency experiments due to the limited bandwidth of the probe. However, when changing the sample or the strength of the magnetic field, the resonant frequency changes. If a passive TR switch is used, it is necessary to frequently replace the $\lambda/4$ line to make the center frequency close to the resonant frequency again to achieve a low insertion loss and high isolation. In addition, for RF pulses of several microseconds, the frequency at the rising edge or the falling edge is far away from the frequency of the middle segment, which may cause high-power leakage to the preamplifier. Compared with passive switches, the active TR switches have wider frequency bands and higher isolation, but those reported in [8], [9], [10], [11], [12], [13], [14], and [15] still cannot work above 500 MHz because of the large insertion loss outside the designed frequency band. For others that can operate above 500 MHz, their insertion loss is large, and their power capacity is less than 100 W (50 dBm), which is generally required in PMF-NMR [16], [17], [18], [19], [20].

WHMFC has achieved the 64 T flat-top pulsed magnetic field [21]. Under the high background field, the resonance frequency of the nuclei is greatly increased. Therefore, the RF switch is required to have a wide frequency band and a high frequency of 1 GHz to almost all cover the resonant frequency of nuclei in materials with special physical properties under high fields that we want to research, such as high-temperature superconducting material, nematic materials and so on. Nuclei with high gyromagnetic ratios such as ^1H and ^9F under high fields are not our research objects for the time being. In addition, due to the short duration of the PMF and the long relaxation time T_1 of the samples, only limited times of experiments can be performed at the top of the magnetic field in high-field PMF-NMR experiments, and the signals cannot even improve the SNR by direct superposition as in steady state field NMR [22]. Therefore, to improve the SNR, the RF switch is required to have a low insertion loss characteristic in its operating frequency band. The insertion loss is less than 0.7 dB in both transmission and reception states, and the signal transmission efficiency can be

guaranteed to be greater than 85%. Moreover, in PMF-NMR, we usually shorten the RF excitation duration and increase the power to reduce the influence of the magnetic field time variation. Therefore, the switch should have a large pulsed power capacity. Finally, the switching time should be as short as possible, e.g., less than 1 μs , to ensure early NMR signal acceptance and improve the SNR [23]. Therefore, it is difficult for the existing RF switches to meet the requirements of a high-frequency broadband, low insertion loss and high power capacity at the same time in PMF-NMR experiments.

In this article, an approximately symmetrical circuit topology of an active RF switch based on PIN diodes and multiple coplanar waveguides with different impedances and its driving circuit are proposed to meet the requirements of a high frequency, broadband, low insertion loss and high-power capacity in PMF-NMR. By setting the insertion loss and return loss optimization goals, we use gradient optimization to determine the optimal gradient direction and tune the circuit parameters and use the least square function to evaluate the error between the optimization goals and the simulation. When the error after optimization is within the acceptable range, we can obtain a set of circuit parameters so that the designed TR switch has excellent performance in the broadband range. Finally, a high-performance TR switch prototype is designed and implemented, which verifies the feasibility of the circuit topology and design method of circuit parameters proposed in this article.

II. HIGH-FREQUENCY BROADBAND TR SWITCH DESIGN METHOD

A. Circuit Design

A schematic of the high-frequency broadband TR switch and its driver circuit is shown in Fig. 3. In Fig. 3(a), the components and impedance lines of the TR switch circuit are symmetrical, except for the Schottky diodes at the RX port. $D1-D4$ are PIN diodes, $Z_1 - Z_6$ and Z_L are the coplanar waveguide lines, and the lines connecting the inductors are all Z_L . $C1-C3$ are used for DC decoupling and cooperate with the inductors to control the impedance. $L2-L5$ provide the DC path for the PIN diodes. $C4$ is the filter capacitor, and the resistor R is the bias current-limiting resistor.

The approximately symmetrical TR switch circuit based on PIN diodes is a series-parallel circuit topology, that is, PIN diodes are connected in series and RF Schottky diodes are connected in parallel. Since PIN diodes in series are used as the switching devices of the RF paths, the TR switch has a wide frequency band [24]. In addition, the parallel RF Schottky diodes can limit the voltage, which is used to further improve the isolation of the TR switch in the transmission state. Moreover, the TR switch circuit uses multiple sections of coplanar waveguide lines and multiple sets of inductors and capacitors, mainly for the purpose of designing the impedance of the TR switch.

Besides inductors and capacitors, both the impedances and the lengths of the coplanar waveguide lines affect the impedance matching of the TR switch. Therefore, we can use the optimization method mentioned in this article to

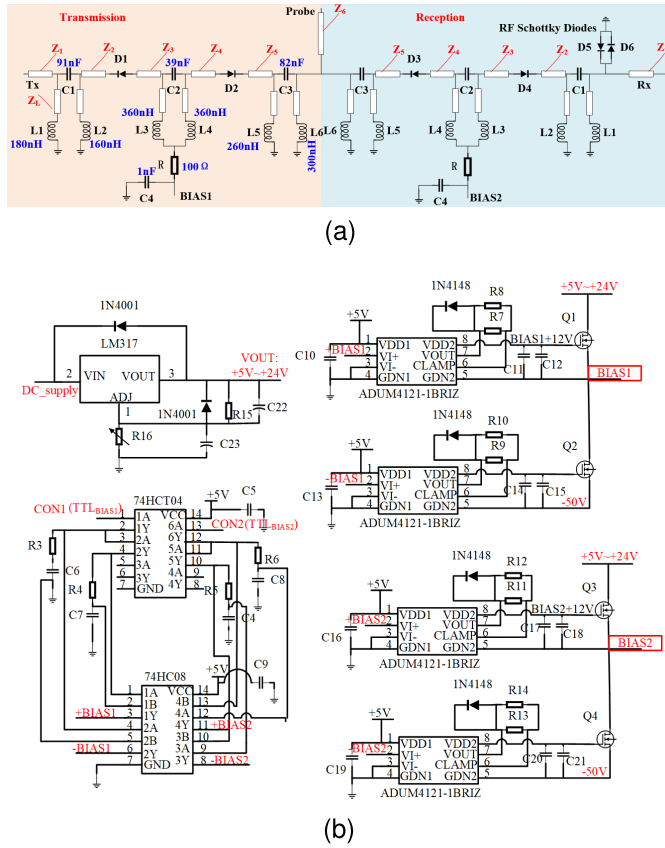


Fig. 3. Schematic of the circuits: (a) TR switch and (b) driver circuit.

adjust the parameters of the coplanar waveguide lines and the values of capacitors and inductors at the same time, instead of just changing the values of capacitors and inductors, so that the impedance of the TR switch can be matched in a wide frequency band. In addition, reasonable adjustment of the impedance and length of the coplanar waveguide can reduce the printed circuit board (PCB) size of the TR switch. Multiple inductors, such as $L1$ and $L2$ rather than only $L1$, can effectively prevent the value of a single inductor from being too large when designing impedance matching, causing its self-resonant frequency (SRF) to be lower than 1 GHz. Moreover, multiple inductors are more conducive to filtering and can effectively reduce the impact of noise caused by the drive circuit.

In Fig. 3(b), the linear voltage regulator LM317 is used to output a continuously adjustable voltage from +5 to +24 V. CON1 (TTL_{BIAS1}) and CON2 (TTL_{BIAS2}) are 3.3 V control signals output by PXI-6452. By using two control signals, we can flexibly set their dead time on the host computer to avoid the influence of RF excitation on the preamplifier when the TR switch switches states. The 74HCT04 is used to convert the control signals from 3.3 to 5 V, and the AND gate logic chip 74HC08 is combined with the RC delay circuits to generate the control signals +BIAS1, -BIAS1, +BIAS2, and -BIAS2, which are used as the inputs of the isolated gate drivers ADUM4121-1BRIZ. RC delay circuits such as $R3$ and $C6$ act as buffers to increase the dead time and prevent D1–D4 from turning on at the same time. The isolated gate drivers output the drive signals that make the nMOS transistors

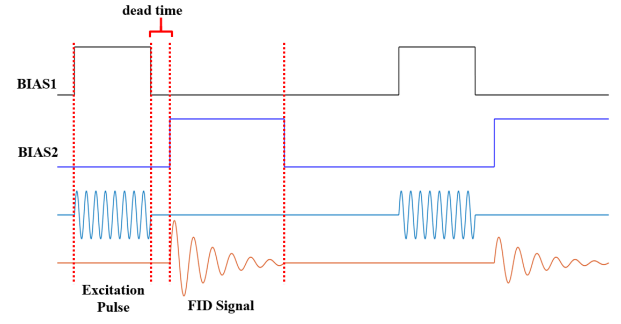


Fig. 4. Timing diagram of a typical one-pulse NMR experiment.

TABLE I
BIAS AND WORKING STATES OF EACH DIODE IN THE TRANSMISSION AND RECEPTION STATES

State	BIAS1	BIAS2	D1-D2	D3-D4	Schottky diodes
Transmission	Positive	Negative	ON	OFF	ON
Reception	Negative	Positive	OFF	ON	OFF

$Q1$ – $Q4$ generate the control signals BIAS1 and BIAS2 for the TR switch.

Fig. 4 shows the working sequence, and Table I displays the bias and working states of each diode. When BIAS1 is high and BIAS2 is low, $D1$ and $D2$ are turned on, $D3$ and $D4$ are turned off, the TR switch works in the transmission state, and the high-power RF pulses are transmitted from the RF pulse input port (Tx) to the sample probe (Probe). To prevent high-power RF pulses from leaking to the low-noise preamplifier, a set of RF Schottky diode pairs is connected in parallel in front of the Rx port to limit the amplitude of voltage and improve the isolation. In contrast, when BIAS1 is low and BIAS2 is high, $D1$ and $D2$ are turned off, $D3$ and $D4$ are turned on, and it is in the state of reception; moreover, the NMR signals received by the probe port are transmitted to the input port (Rx) of the low-noise preamplifier.

To improve the SNR, PMF-NMR experiments require the TR switch to operate in a low insertion loss state. The parameters of the impedance lines and the values of the inductors and capacitors will affect the impedance matching of the TR switch in the broadband range and thus affect the insertion loss and return loss of the TR switch. Therefore, we can design the parameters of the waveguide lines and the values of each inductor and capacitor to make the TR switch maintain a high return loss and low insertion loss in the broadband range to ensure that the weak NMR signals can be transmitted to the preamplifier more efficiently. Moreover, large-package PIN diodes with heatsinks, such as MACOM's 1072-size package series, are sufficient to meet the power capacity requirement of 200 W RF pulses.

In addition, since the circuit is approximately symmetrical and the Schottky diodes are turned off when the weak NMR signals (μV level) are received, the components and paths of the RF pulses to the probe and the NMR signals to the Rx are exactly the same in both the transmission and reception states.

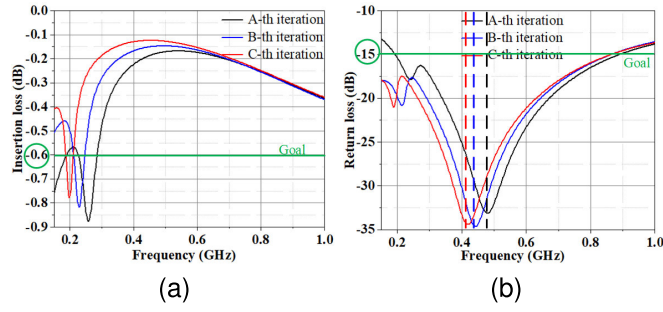


Fig. 5. Three optimization iterations: (a) insertion loss and (b) return loss.

Therefore, as long as the insertion loss under one working state is guaranteed, the insertion loss in the other state is almost the same, which simplifies the design process of the TR switch.

B. Impedance Design of Transmission Lines and Device Selection of the TR Switch

The impedance parameters of waveguide lines and the values of inductors and capacitors are obtained by optimizing the return loss and the insertion loss based on the gradient optimizer and the least-squares error function (EF) [25], [26]. As shown in (1), two optimization goals of the insertion loss $S(\text{Probe}, \text{Tx})$ and return loss $S(\text{Tx}, \text{Tx})$ at 250 MHz–1 GHz are set. Then, the initial parameters of the waveguide lines and the values of the inductors and capacitors are set. The gradient optimizer uses the gradient search method to arrive at new parameters and values by using the gradient information of the EF of the network. The formula of EF is also shown in (1), and W_i is the weight of the objective function; here, we take $W_{S(\text{Probe}, \text{Tx})} = 2$ and $W_{S(\text{Tx}, \text{Tx})} = 1$, which indicates that we prefer to reduce the insertion loss error. simulation_i and goal_i are the simulation results and goals for the insertion loss and return loss, respectively. The least-squares EF is calculated by evaluating the error for each specified goal at each frequency point individually. The EF and its gradient first evaluate the parameters of the waveguide lines and the values of inductors and capacitors and determine the direction in which a set of parameters and values needs to be moved to reduce the error. Once the direction is determined, the set of the parameters of the waveguide lines and the values of the inductors and capacitors are moved in that direction until the EF is minimized. Then, the gradient is reevaluated. After many iterations, we can obtain a set of the impedance parameters of the waveguide lines and the values of the inductors and capacitors that make the EF within an acceptable range

$$\begin{cases} S(\text{Probe}, \text{Tx}) > -0.6 \text{ dB} \\ S(\text{Tx}, \text{Tx}) < -15 \text{ dB} \\ \text{EF} = \sum_{\text{goals}} W_i \times |\text{simulation}_i - \text{goal}_i|^2 \\ W_{S(\text{Probe}, \text{Tx})} = 2 \\ W_{S(\text{Tx}, \text{Tx})} = 1. \end{cases} \quad (1)$$

Fig. 5 shows a set of data in the optimization process. During the optimization process from the A-th to the B-th to the C-th time, the insertion loss error decreases and gradually

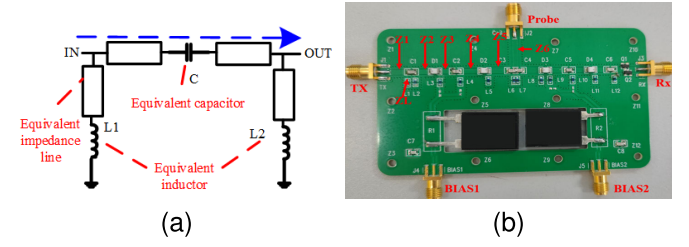


Fig. 6. (a) Equivalent circuit when the TR switch is unidirectionally conducting. (b) PCB photograph of the TR switch.

TABLE II
IMPEDANCE VALUES AND THE LENGTHS OF EACH LINE

Impedance line number	Impedance value / Ω	Length/mm
Z_1	46.3	10.0
Z_2	54.0	4.0
Z_3	54.0	4.0
Z_4	56.3	4.0
Z_5	72.7	3.5
Z_6	66.4	1.5
Z_L	93.0	4.5

approaches the set goal values. Fig. 5(b) clearly shows that the center frequency of the matching is shifted to the left. As shown in Fig. 6(a), the TR switch can be equivalent to an LC filter circuit when the TR switch is unidirectionally conducting and the gradient optimization direction is increasing the equivalent parallel inductances and series capacitances in the TR switch circuit. By utilizing the gradient optimization method, we obtain the designs of each coplanar waveguide line, as shown in Table II, and the values of each inductor and capacitor.

Considering the actual values of the capacitor and inductor and the SRF that needs to be far away from the operating frequency band of the TR switch, we use the Murata LQW18AN series RF inductors and Johanson 251R15S series RF capacitors whose SRF is above 1 GHz, and the values of each inductor and capacitor are shown in Fig. 3(a). Since the $\lambda/4$ line is not used, the size of the TR switch can be reduced to only 6*10 cm, and the photograph of the TR switch whose PCB prototype is Rogers 4350B is shown in Fig. 6(b).

In the TR switch, the PIN diodes MA4P504-1072T selected are the core devices with an operating frequency range of 1 MHz–4 GHz, a maximum reverse voltage of 500 V, a maximum equivalent series resistance of 0.6 Ω , a carrier lifetime of 1 μs , and a maximum continuous wave dissipation power of 7.5 W [27]. Within the working frequency range, the PIN diode can meet the requirements of high frequency and broadband. Equation(2) gives the power capacity of the PIN diode in series, where R_S is the equivalent series resistance of the PIN diode, and Z_0 is the matching impedance of 50 Ω . When the available power P_{AV} is 200 W, the dissipation power of the PIN diode is $P_d = 2.37$ W, which is less than 7.5 W. In NMR experiments, the RF excitation is the pulsed mode. Therefore, the power capacity of the diode can meet the requirements of transmitting high-power RF pulses in the

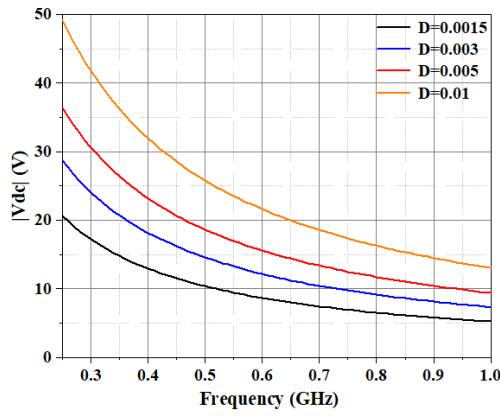


Fig. 7. Relationship between the minimum reverse bias voltage and the duty cycle under 200 W RF pulses for PIN diode MA4P504-1072T.

PMF-NMR experiments

$$P_d = \frac{4R_S Z_0}{(2Z_0 + R_S)^2} P_{AV}. \quad (2)$$

The off-time of the PIN diode is given by (3). I_F is the forward bias current, τ is the carrier lifetime, and I_R is the reverse bias current. When I_F is 100 mA and I_R is 200 mA, the calculated T_{FR} is 0.405 μ s. Considering the dead time set on the host computer and the delay time brought by the PIN diode drive circuit, the total switching time is still less than 1 μ s. Since the reverse turn-off time of the PIN diode is much longer than the forward turn-on time [24], this switching speed meets the requirement of fast switching in PMF-NMR experiments

$$T_{FR} = \ln(1 + I_F/I_R)\tau. \quad (3)$$

It can be seen from Fig. 3(b) that BIAS1 and BIAS2 can output a reverse bias voltage of -50 V. Due to the short duration of the pulsed field, the duty cycle of the RF excitation used is generally less than 0.01; for example, the period is 1 ms, and the RF pulse excitation lasts 10 μ s. According to the relationship between the minimum reverse bias voltage of the PIN diode and parameters such as the RF excitation duty cycle, carrier lifetime of the PIN diode, and RF pulse power [28], we can obtain the curve shown in Fig. 7. Therefore, the designed drive circuit can maintain the PIN diode MA4P504-1072T to be reverse-biased under 200 W RF pulses.

A pair of Schottky diodes BAT15-04R with a small junction capacitance, low forward voltage, and extremely short reverse recovery time is connected in parallel to the Rx port. It can limit the voltage amplitude to further protect the preamplifier behind the Rx port from voltage transients and improve isolation when the switch state is switched.

C. S-Parameters Simulation

The analog circuit simulation software used is Advance Design System (ADS). After the selections of the inductors and capacitors are finally determined, their official models are used in the simulation, and the simulation results of the S-parameters are shown in Fig. 8. The TR switch is designed to operate at 250 MHz–1 GHz. The return loss in both states

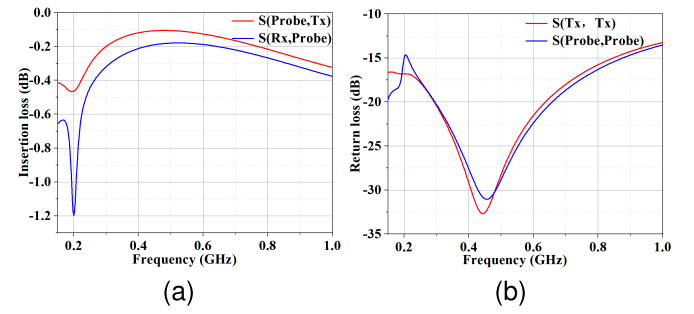


Fig. 8. S-parameter simulation of the TR switch (red: transmission state, blue: reception state): (a) insertion loss and (b) return loss.

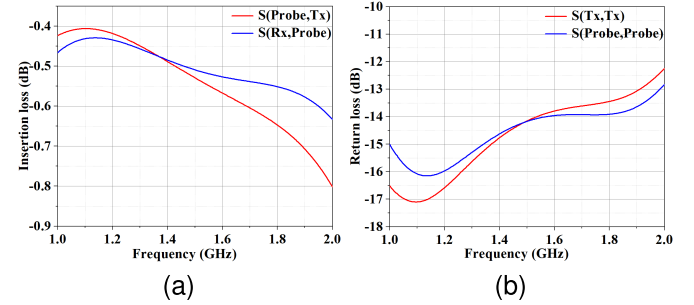


Fig. 9. Simulation of the TR switch designed by the method proposed in the 1–2 GHz frequency band: (a) insertion loss and (b) return loss.

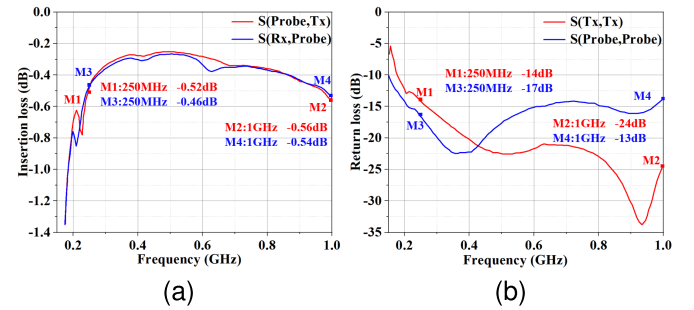


Fig. 10. TR switch characterization results: (a) insertion loss and (b) return loss.

is greater than 13 dB, which indicates that only 5.3% of the power is reflected back to the Tx, and the insertion loss is less than 0.5 dB. This causes the transmission efficiency to be greater than 89%.

Similarly, we can use the above design method to meet the needs of higher-frequency PMF-NMR experiments by only adjusting the impedance parameters of the waveguide lines and the values of the inductors and capacitors while maintaining an unchanged TR switching topology. Fig. 9 shows the simulation of the insertion loss and return loss of the TR switch at 1–2 GHz.

III. MEASUREMENTS AND DISCUSSION

A. Measurement of the Insertion Loss and Return Loss

Since the insertion loss and return loss are only related to the states of the switch and the frequency of the signal, the network analyzer ZNB20 is used to test the insertion loss and return loss, and the test results are shown in Fig. 10. In the frequency range of 250 MHz–1 GHz, the insertion loss of

TABLE III

COMPARISON BETWEEN THE SIMULATIONS AND ACTUAL TESTS OF THE INSERTION LOSS AND RETURN LOSS IN THE 250 MHz–1 GHz FREQUENCY BAND

State	Simulation or Test	Insertion loss (dB)	Efficiency (%)	Return loss (dB)
TX	Simulation	0.125-0.4	91-97	>13
TX	Test	0.25-0.56	88-94	>14
RX	Simulation	0.2-0.5	89-95	>13
RX	Test	0.25-0.54	88-94	>13

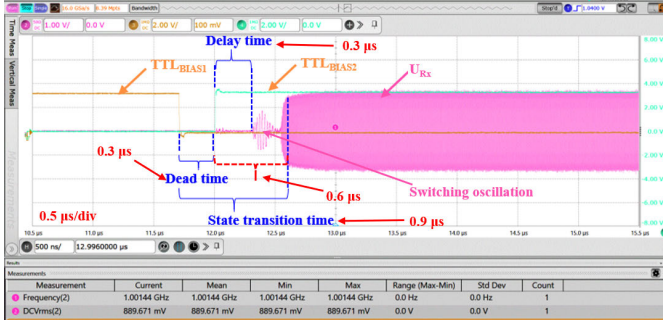
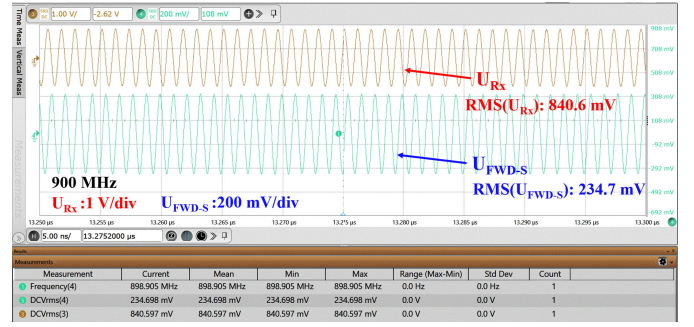


Fig. 11. Test of state transition time of the TR switch.

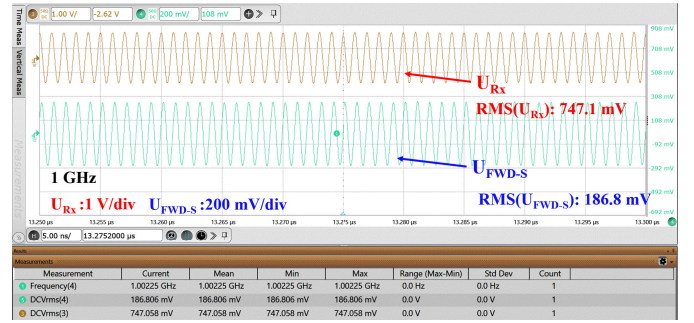
the TR switch in the two states is less than 0.56 dB, which indicates the minimum transmission efficiency is 88%. The return loss in both states is greater than 13 dB, and the voltage standing wave ratio (VSWR) is less than 1.6, which is the same as that of the simulation. Table III shows that the test results of the insertion loss and return loss are very close to the simulation results. The transmission efficiency of the signals is only 3% error. Therefore, this also confirms the correctness of the simulation and design methods to a large extent.

B. Measurement of the State Switching Time

In addition, the TR switch should be able to quickly switch so that the NMR signal at the beginning of the decay can be recorded when switching from the transmitting mode to the receiving mode. Therefore, the state switching time of the TR switch is tested, as shown in Fig. 11. We set the dead time to 0.3 μ s and add an RF signal to the probe port after TTL_{BIAS2} is high at 0.55 μ s to verify that the TR switch transitions from the transmit state to the receiving state. The delay time between TTL_{BIAS2} and the bias voltage $BIAS2$ generated by the PIN diode drive circuit is approximately 0.3 μ s. After TTL_{BIAS1} has been at a low level of 0.9 μ s, the voltage envelope of Rx returns to about 95% of the stable value and the RF signal is transmitted from the probe to the Rx without distortion, which indicates that the TR switch completely changes from the transmission state to the reception state. That is, the T_{FR} of the PIN diodes is less than 0.6 μ s, and the switching time is within 0.6 μ s. Therefore, the state transition time, including the dead time and the delay time brought by the drive circuit, is less than 0.9 μ s, which meets the requirements of fast switching.



(a)



(b)

Fig. 12. Isolation test (brown curve: U_{Rx} , green curve: U_{FWD-S}): (a) 900 MHz 176 W and (b) 1 GHz 110 W.

C. Measurement of the Power Capacity and Isolation

The power capacity and isolation are two closely related parameters of the TR switch. The higher the power capacity is, the better the isolation required to ensure that the preamplifier behind the Rx port will not be damaged. Therefore, more attention is given to isolation under high-power signals rather than isolation under small signals. The forward power sample is the signal attenuated by 52 dB from the output power of the RF amplifier used. Thus, the voltage of the RF excitations is 400 times the voltage of the forward power sampling port (U_{FWD-S}). We use the relationship between U_{FWD-S} and the voltage of the Rx port (U_{Rx}) to test the isolation of the TR switch. However, the maximum power is limited by the PA, which can only output 200 W of RF pulses and can only output 176 W at 900 MHz, and only 110 W at 1 GHz. As shown in Fig. 12, the leakage power of the Rx port is 14 mW (11.46 dBm) and 11.16 mW (10.48 dBm) when RF pulses of 176 W (52.46 dBm) at 900 MHz and 110 W (50.4 dBm) at 1 GHz are applied, respectively.

From the comparison of Figs. 12 and 13(a), it can be seen that the test results are relatively close to the isolation of the simulation, which makes it possible to use the simulation to illustrate the isolation under high-frequency and high-power RF pulses. Since the RF Schottky diodes are not saturated when the 900 MHz 176 W RF pulse is applied, and the series-connected PIN diodes MA4P504-1072 only reduce the isolation by about 1 dB when the operating frequency increases from 900 MHz to 1 GHz [27], the isolation of the TR switch is still close to 40 dB under 1 GHz 200 W RF pulses. Therefore, we replace the test results of 900 MHz–1 GHz

TABLE IV
COMPARISON WITH THE REPORTED STATE-OF-THE-ART RF TR SWITCHES

Reference	Year	Freq range (MHz)	Insertion loss (dB)	Isolation (dB)	Return loss (dB)	Power capacity (W)	Switching time (μ s)
This work	2022	250 - 1000	0.2 - 0.56	> 40	> 13	200	0.9
[8]	2021	30 - 200	0.09 - 0.37	63-80	> 18	125 (CW)	< 1.9
[9]	2021	100 - 500	0.3 - 3	5 - 45	> 30	1000	NR
[19]	2014	100-1500	0.475 - 0.9	NR	> 27	0.21	NR

CW: continuous wave. NR: not reported.

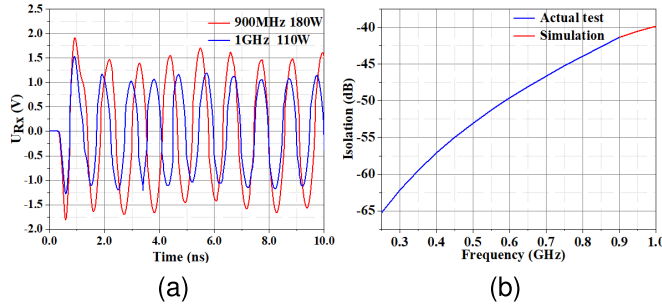


Fig. 13. (a) Simulation of the leakage voltage of the Rx port. (b) Isolation of the Tx and Rx ports under a 200 W RF pulse (900 MHz–1 GHz is the simulation).

with simulated isolation, and the others are actual tests under approximately 200 W RF excitation. The curve fitting of the isolation test under a 200 W RF pulse is shown in Fig. 13(b), and the isolation of the TR switch is larger than or equal to 40 dB.

Regarding common preamplifiers, the short-term maximum input power can reach 30 dBm [29]. Therefore, the isolation of the Tx and Rx ports of the TR switch can meet the requirements and will not affect the normal operation of the preamplifier. For the receive mode, we care more about the insertion loss than the isolation loss because the NMR signal is very weak.

D. PMF-NMR Experiment of ^{93}Nb

To verify the performance of the TR switch, the PMF-NMR experiments are performed on ^{93}Nb nuclei under a 39.55 T flat-top PMF. Equation (4) is the calculation formula of the resonance frequency, where γ is the gyromagnetic ratio, K is the Knight shift, and B_0 is the background magnetic field. For ^{93}Nb , $\gamma = 10.405$ MHz/T, $K = 0.87\%$, and the resonance frequency $f = 415$ MHz under 39.55 T

$$f = \gamma(1 + K)B_0. \quad (4)$$

We used a train of 9 RF pulses (pulsewidth $\tau = 9 \mu\text{s}$, period $T = 1$ ms, power $P = 50$ W, $f = 415$ MHz) to excite the ^{93}Nb nuclei. A total of nine sets of free induction decay (FID) signals of ^{93}Nb are obtained, and one of them is shown in Fig. 14. The experiment verifies that the TR switch designed in this article can be used in PMF-NMR experiments, and it can be seen from Fig. 14(a) that the state transition time of the TR switch is $0.9 \mu\text{s}$. Moreover, compared with the noise

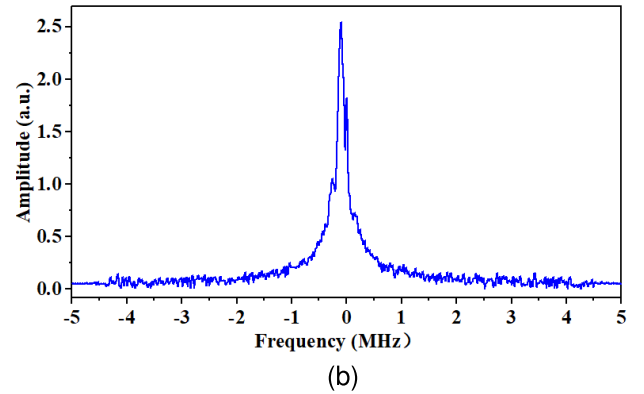
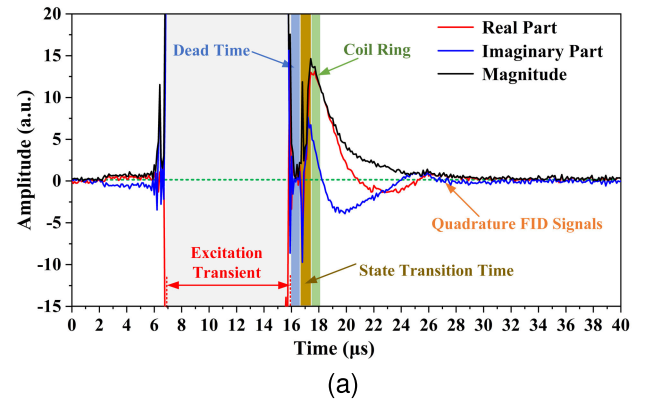


Fig. 14. Time-domain and frequency-domain diagrams of FID signals of ^{93}Nb under 39.55 T: (a) FID signals of ^{93}Nb after quadrature down-conversion and (b) corresponding NMR spectra of ^{93}Nb at the excited RF of 415 MHz.

signals when the TR switch is not working within $0\text{--}7 \mu\text{s}$, it can be seen that the TR switch introduces almost no noise signals, and FID signals have a high SNR, which indicates that noise signals of the drive circuit have little interference on PMF-NMR FID signals after the filtering effect of multiple inductors of the TR switch circuit.

E. Comparison of the Performance Parameters

The performance of the proposed switch is compared with the state-of-the-art TR switch designs reported in Table IV. Compared with other designs reported, the TR switch proposed in this article has excellent performance. The designed TR switch has a wide frequency band of 250 MHz–1 GHz. Compared with [8], its high-frequency and broadband performance are more advantageous in PMF-NMR applications. In the

transmit and receive modes, its insertion loss is also very small, both less than 0.56 dB, and the power transmission efficiency is greater than 88%. In [9], the maximum insertion loss is approximately 3 dB in the 100–500 MHz frequency band. In [30], a 1 kW high-power T/R switch for MRI was designed for 14.2 T to 600 MHz with over 60 dB isolation, but also up to 1 dB insertion loss in transmission and reception states. Moreover, the power capacity of 200 W is larger than that of [19], which is only 0.21 W for the same frequency. The isolation of Tx-Rx can reach 40 dB under the RF pulse of 1 GHz 200 W, which is smaller than those of the references. However, after the calculation and analysis, it can also be used without disturbing the normal operation of the preamplifier. The state switching time of less than 1 μ s also makes it easier to receive the initial NMR signals compared to the switching time of 1.9 μ s in [8]. The TR switch designed in this article achieves excellent performance in the broadband range, while other switches have only observed comparable performance in a few cases.

IV. CONCLUSION

In this article, an active TR switch topology is proposed based on PIN diodes and multiple sections of coplanar waveguide lines with different impedances to meet the requirements of the TR switch for a high-frequency broadband, low insertion loss, and high-power capacity in PMF-NMR experiments. A high-performance TR switch prototype with a frequency band of 250 MHz–1 GHz, insertion loss less than 0.56 dB, switching time less than 0.9 μ s, and isolation higher than 40 dB under a 200 W RF pulse is designed and implemented based on this topology. The active TR switch avoids the narrow-band characteristic of the $\lambda/4$ lines. Compared with other existing active TR switches, these excellent properties allow the active TR switch designed in this article to be applied to the resonance frequencies of almost all nuclei under a pulsed high magnetic field. In addition, to meet the high-field PMF-NMR experimental requirements of nuclei with high gyromagnetic ratios such as ^1H and ^9F , the operating frequency of the TR switch can be further increased to above 1 GHz by using this approximately symmetrical TR switch circuit topology and the gradient optimization mentioned in this article.

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